Failure of the sharp affine-modulus bound

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Abstract

A small explicit harmonic perturbation of the critical circular-annulus map violates the proposed bound even after optimizing over all real affine transformations. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

Catalogue scope. AIM 689; complete resolution. Canonical target at revision `1e56463`.

1 Statement and argument

The sharp necessary lower bound proposed in [1, Conjecture 1.3] fails under a small explicit harmonic perturbation of the critical circular-annulus map. The affine optimization is included below; decreasing ordinary modulus alone would not settle the question.

Theorem 1.1. *Fix $\tau > 0$ and put $R = \cosh\tau$. For every sufficiently small nonzero real ε, the harmonic map*

$$h_\varepsilon(z) = \frac{1}{2}\left(z + \frac{1}{\bar z}\right) + \frac{\varepsilon}{2}\left(z^2 + \frac{1}{\bar z^2}\right), \qquad 1 < |z| < e^\tau, \tag{1}$$

is a homeomorphism onto a bounded ring Ω_ε with analytic boundary. Its target satisfies

$$\mathrm{Mod}_{\mathrm{aff}}(\Omega_\varepsilon) < \log R = \log\cosh(\mathrm{Mod}\{1 < |z| < e^\tau\})\,.$$

Thus the inequality in Problem 689 is false.

Proof. Write

$$S = \cosh(2\tau) = 2R^2 - 1, \qquad m = \log R > 0.$$

In logarithmic polar coordinates,

$$h_\varepsilon(e^{t+i\theta}) = \cosh t\, e^{i\theta} + \varepsilon\cosh(2t)\, e^{2i\theta}. \tag{2}$$

Its two boundary curves are

$$\Gamma_{0,\varepsilon}(\theta) = e^{i\theta} + \varepsilon e^{2i\theta}, \qquad \Gamma_{1,\varepsilon}(\theta) = Re^{i\theta} + \varepsilon S e^{2i\theta}. \tag{3}$$

For sufficiently small $|\varepsilon|$, these are analytic Jordan curves. They are nested, for example, whenever $|\varepsilon| < (R-1)/(S+1)$, since the inner curve lies in the disk of radius $1 + |\varepsilon|$ while the outer curve encloses the disk of radius $R - |\varepsilon|S$. Let Ω_ε be the ring between them.

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

Homeomorphism. The Jacobian of (2) with respect to (t, θ) is

$$J_\varepsilon(t,\theta) = \frac{1}{2}\sinh(2t) + 2\varepsilon\sinh(3t)\cos\theta + 2\varepsilon^2\sinh(4t). \tag{4}$$

Because $\sinh(3t)/\sinh(2t)$ extends continuously to $t = 0$ and is bounded on $[0, \tau]$, this is positive for $0 < t \le \tau$ when $|\varepsilon|$ is small enough. Each intermediate curve is a Jordan curve: the polynomial

$$w \longmapsto \cosh t\, w + \varepsilon\cosh(2t)w^2$$

is injective on the closed unit disk if $2|\varepsilon|\cosh(2t) < \cosh t$, which holds uniformly for $0 \le t \le \tau$ after reducing $|\varepsilon|$. The boundary winding degree is one between (3), and zero outside. Every interior preimage has positive Jacobian by (4). The degree theorem therefore gives exactly one preimage for every point between the boundary curves, and none outside. Local invertibility proves that h_ε is a homeomorphism onto Ω_ε. Each summand in (1) is holomorphic or antiholomorphic, so the map is harmonic.

Second-order modulus formula. We record a short boundary-perturbation calculation used twice below. Work on the logarithmic cylinder with coordinate $s + i\phi$, of imaginary period 2π. Suppose the boundaries of a perturbed ring are

$$s = s_j + \delta p_j\cos(k\phi) + \delta^2(q_j + \text{zero-mean Fourier terms}) + O(\delta^3), \quad s_0 = 0,\ s_1 = m. \tag{5}$$

The remainder is uniform with its first two derivatives. Let A, B solve

$$A + B = p_0, \qquad Ae^{km} + Be^{-km} = p_1, \qquad t_j = Ae^{ks_j} - Be^{-ks_j}. \tag{6}$$

Then its modulus is

$$m + \delta^2\left[q_1 - q_0 - \frac{k}{2}(p_1t_1 - p_0t_0)\right] + O(\delta^3). \tag{7}$$

To verify this, use the periodic conformal approximation

$$W(\zeta) = \zeta + \delta(Ae^{k\zeta} + Be^{-k\zeta}) + \delta^2K(\zeta).$$

On the jth source boundary the angular variable becomes $\phi = y + \delta t_j\sin(ky) + O(\delta^2)$. Substitution into (5) makes the mean of the real boundary coordinate

$$s_j + \delta^2\left(q_j - \frac{k}{2}p_jt_j\right) + O(\delta^3).$$

The mean of $\Re K$ is the same on both straight boundaries. Hence their separation changes by the difference of these two second-order means, proving the coefficient in (7).

Here the calculation can be justified without assuming an unproved smooth dependence of uniformization. The second-order boundary discrepancies are finite Fourier sums. For each nonzero Fourier mode, two periodic exponential terms in K match the two boundary discrepancies; the relevant two-by-two system is invertible since $m > 0$. The two mean discrepancies determine the source-circle shifts. The resulting finite-Laurent map is injective for small δ, since it is C^1-close to the identity on a slightly larger closed annulus. Its boundary mismatch is $O(\delta^3)$. Moving the source boundaries inward and outward by $C|\delta|^3$, for a sufficiently large fixed C, sandwiches the desired ring between two such conformal images. Modulus monotonicity bounds the error by $O(\delta^3)$. The same construction with finitely many real parameters gives a joint quadratic Taylor polynomial with a uniform cubic remainder: the boundary data and the Fourier correction coefficients depend polynomially on those parameters to this order.

Joint analytic dependence. There is also a fixed-domain justification of the uniform Taylor remainder used below. On $A_R = \{1 < |w| < R\}$ the smooth map

$$F_\varepsilon(w) = w + \varepsilon(2|w|^2 - 1)(w/|w|)^2$$

maps both boundary circles exactly to (3) and is a diffeomorphism for small complex ε. For small complex β, the composition $L_\beta \circ F_\varepsilon$, where $L_\beta(z) = z + \beta\bar{z}$, is a real-analytic parameter family of diffeomorphisms. Pulling back the condenser energy gives a uniformly elliptic quadratic energy on the fixed annulus, with matrix coefficients real-analytic in the four real parameters. After fixing a smooth lift of the boundary values, Lax–Milgram and the Neumann series for the perturbed inverse give real-analytic dependence of the H^1_0 minimizer and its energy. The modulus is 2π divided by this positive capacity, and is therefore jointly real-analytic as well. This supplies the uniform cubic Taylor remainder independently of the finite-Laurent construction.

The harmonic perturbation decreases modulus. For a curve $z = re^{i\theta} + \varepsilon c e^{2i\theta}$, put $d = c/r$. Expanding in the physical angular coordinate $\phi = \arg z$ yields

$$\log|z| = \log r + \varepsilon d\cos\phi + \varepsilon^2\left(\frac{d^2}{2} - d^2\cos 2\phi\right) + O(\varepsilon^3). \tag{8}$$

Indeed, $\phi = \theta + \varepsilon d\sin\theta + O(\varepsilon^2)$; substituting its inverse into $\Re\log(1 + \varepsilon d e^{i\theta})$ gives the displayed formula. For the two curves in (3), use $k = 1$, $p_0 = 1$, $p_1 = S/R$, and $q_j = p_j^2/2$ in (7). The constants in (6) are $A = 2$ and $B = -1$. It follows that

$$\mathrm{Mod}(\Omega_\varepsilon) = m - \left(1 - R^{-2}\right)\varepsilon^2 + O(\varepsilon^3). \tag{9}$$

Local affine optimization. After composition with a similarity and, if needed, conjugation, any invertible real affine map relevant to modulus is represented by

$$L_\beta(z) = z + \beta\bar{z}, \qquad |\beta| < 1.$$

Translations and similarities do not affect modulus. Set

$$M(\varepsilon, \beta) = \mathrm{Mod}(L_\beta\Omega_\varepsilon).$$

First take $\varepsilon = 0$ and $\beta = b$ real. On the image of a circle,

$$\log|r(e^{i\theta} + be^{-i\theta})| = \log r + b\cos 2\phi + b^2\left(-1 + \frac{1}{2}\cos 4\phi\right) + O(b^3). \tag{10}$$

Now (7) applies with $k = 2$, $p_0 = p_1 = 1$, $q_0 = q_1 = -1$, and

$$A = \frac{1}{R^2 + 1}, \qquad B = \frac{R^2}{R^2 + 1}.$$

Consequently,

$$M(0, b) = m - 2\frac{R^2 - 1}{R^2 + 1}b^2 + O(b^3). \tag{11}$$

Rotational invariance makes the complex-β quadratic term a multiple of $|\beta|^2$ with the same coefficient.

There are no mixed quadratic terms between the harmonic and affine perturbations. To see this, temporarily allow ε to be complex in the boundary parametrizations. Rotation by an angle α and boundary reparametrization give

$$M(\varepsilon, \beta) = M(e^{-i\alpha}\varepsilon, e^{2i\alpha}\beta). \tag{12}$$

A quadratic monomial with one factor from $\varepsilon, \bar{\varepsilon}$ and one from $\beta, \bar{\beta}$ cannot be invariant under this action. Equivalently, the first-order perturbations have angular frequencies one and two, whose

pairwise products have no constant Fourier coefficient. The joint Taylor construction above, together with (9) and (11), therefore yields

$$M(\varepsilon,\beta) = m - q|\varepsilon|^2 - a|\beta|^2 + O((|\varepsilon|+|\beta|)^3), \qquad q = 1 - R^{-2} > 0, \quad a = 2\frac{R^2-1}{R^2+1} > 0. \tag{13}$$

In a fixed neighborhood of $(0,0)$ this gives

$$M(\varepsilon,\beta) \le m - \frac{q}{2}|\varepsilon|^2 - \frac{a}{2}|\beta|^2. \tag{14}$$

Excluding affine maps far from the identity. For a smooth ring whose two boundary curves enclose areas $A_0 < A_1$, Carleman's area inequality and its equality case are

$$\mathrm{Mod}(\Omega) \le \frac{1}{2}\log(A_1/A_0), \quad \text{with equality only for a circular annulus up to similarity.} \tag{15}$$

The needed version has a short proof. A conformal parametrization $f(w) = \sum_{k\in\mathbb{Z}} a_k w^k$ from $1 < |w| < e^M$, taking inner boundary to inner boundary, satisfies

$$A(r) = \pi \sum_{k\in\mathbb{Z}} k|a_k|^2 r^{2k}.$$

For analytic Jordan boundaries the map extends across the boundaries, so this area identity and its boundary limits are valid. Then

$$A(e^M) - e^{2M}A(1) = \pi \sum_{k} k|a_k|^2 (e^{2kM} - e^{2M}) \ge 0.$$

All nonzero summands are positive except those for $k = 0, 1$. Equality forces $f(w) = a_0 + a_1 w$, proving (15) with its equality case. For the image of the round annulus under L_β, the enclosed-area ratio remains R^2, since the affine determinant cancels. If $\beta \neq 0$ the boundary curves are noncircular ellipses. Thus

$$M(0,\beta) < m \qquad (\beta \neq 0). \tag{16}$$

On a compact set of parameters bounded away from both 0 and the unit circle, this strict deficit is uniform and persists for sufficiently small ε by continuity under smooth boundary perturbations. That continuity also follows from the fixed-domain Dirichlet-energy formulation.

It remains to control $|\beta| \uparrow 1$ uniformly. The bounded inner complementary component of Ω_ε contains the disk of radius $1 - |\varepsilon|$, while its outer boundary lies inside the disk of radius $R + |\varepsilon|S$. Hence, by ring-modulus monotonicity,

$$M(\varepsilon,\beta) \le \mathrm{Mod}\left(L_\beta\{1 - |\varepsilon| < |z| < R + |\varepsilon|S\}\right). \tag{17}$$

Up to rotation, L_β has axes $1 + |\beta|$ and $1 - |\beta|$. For the elliptical ring obtained from $r < |z| < s$, every horizontal coordinate $|x| \le (1 + |\beta|)r/2$ determines an upper vertical segment joining the two boundary ellipses. Its length is at most $(1 - |\beta|)s$. Every admissible condenser potential changes by one on this segment. Cauchy–Schwarz and integration in x bound its Dirichlet energy below by

$$\frac{(1+|\beta|)r}{(1-|\beta|)s}.$$

Since ring modulus is 2π divided by condenser capacity, the modulus is at most

$$2\pi\frac{(1-|\beta|)s}{(1+|\beta|)r}.$$

For the radii in (17), r stays bounded below and s bounded above as $\varepsilon \to 0$. Thus $M(\varepsilon,\beta) \to 0$ uniformly as $|\beta| \uparrow 1$.

Choose a small parameter neighborhood in which (14) holds. The compact strict bound from (16) and the degeneration estimate show that all remaining affine parameters have modulus at most $m - \gamma$ for some fixed $\gamma > 0$, for sufficiently small $|\varepsilon|$. Shrinking $|\varepsilon|$ further if necessary, the full affine supremum is bounded by

$$\mathrm{Mod}_{\mathrm{aff}}(\Omega_\varepsilon) \le m - \frac{q}{2}\varepsilon^2 < m \qquad (\varepsilon \neq 0).$$

Together with the harmonic homeomorphism (1), this proves the theorem. □

Independent numerical check. The proof is asymptotic and does not depend on numerical optimization. As a cross-check, at $\tau = 1$ the baseline is $m = 0.4337808304830271$. Harmonic Laurent collocation gives $(\mathrm{Mod}(\Omega_\varepsilon) - m)/\varepsilon^2$ equal to -0.5841929, -0.5810615, -0.5802843, and -0.5800670 at $\varepsilon = 0.02, 0.01, 0.005, 0.002$, respectively, approaching the proved coefficient $-(1 - R^{-2}) = -0.5800256584$. At $\varepsilon = 0.02$, numerical affine optimization gives $\beta \simeq 0.003933496$ and modulus 0.433559780729, still below the proposed bound. These values provide a consistency check only.

References

[1] T. Iwaniec, L. V. Kovalev and J. Onninen, *Harmonic mapping problem and affine capacity*, Proc. Roy. Soc. Edinburgh Sect. A **141** (2011), 1017–1030. https://arxiv.org/abs/1001.2124.